

Hub integration: finishing the three Hubs on the new architecture

A data-rebind and validation plan to move the Data, Modeling, and Experimentation Hubs from the original S3-fixture design onto live NEON input and in-container CESM runs.

Data Science Institute, UW-Madison · July 10, 2026

EFFORT

~5.5 eng-days

SITES

5 NEON towers

MODEL RUNS

Proven (KONZ)

OWNER

Steve / Abe

Executive summary

The Hubs, analytics backend, and perturbation tooling all exist, and CTSM runs to completion in the container. Finishing the work is a **data-rebind and validation** effort, not a redesign: every Hub reads from a hardcoded UW-Madison S3 store and must be repointed at the live/native data flow, then validated against reference outputs.

Now unblocked. The July 9 comment thread on `communication-internal.docx` settled the open design questions: move to live/native data, the Experimentation Hub loop is confirmed, and the Kalman filter target is set.

What is now decided

QUESTION	RESOLUTION
S3 fixtures vs. live/native data	Go native/live. S3 becomes an optional sample copy, not a requirement.
Experimentation Hub architecture	Confirmed 4-step loop: one NEON input → one perturbed copy → run CESM on both → t-test the two outputs.
Kalman filter target parameter	GPP (Gross Primary Productivity).
Sample sites	5 stations , with portable copies in a shared Google Drive folder.
Ownership	Reassigned to Steve; Maria is now a reference.

Deliverable

A self-contained **local Docker image**, single-user, **no auth**, in which all three Hubs run end-to-end against **live NEON input and in-container CESM output**, validated on the five sample sites.

Out of scope (follow-on): VM hosting, multi-user, authentication, and S3 credential-rotation security work.

What the rebind actually involves

Revised 2026-07-30. This section previously said the two layouts shared one filename convention. Phase 0 compared them against a real KONZ run and found they do not: the current readers match **zero** live files. The rebind is still contained, but it is not a pure prefix swap.

```
S3 / reference : archive_1/{site}.transient/lnd/hist/{site}.transient.clm2.h1.YYYY-MM-DD-000000.nc
Live run      : {output_root}/archive/lnd/hist/{site}.transient.clm2.h1a.YYYY-MM-DD-01800.nc
```

Three differences, each of which breaks the current filter: the stream token gains a suffix (`h1` → `h1a` , `h0` → `h0a`), the per-site subdirectory disappears, and the daily seconds-of-day field is not `00000` . The reference copies keep the *legacy* naming, so the reader must handle both conventions rather than switching to the new one. Two files need the fix, not one: `analytics_modules/data_access.py` and `cesm-tools/site_and_regional/run_neon_v2.py` .

Still true: the soil-profile plotting function already has a local-path branch, and the core change remains flipping the default source and generalizing the two S3-only readers.

Roadmap

PHASE 0 Data contract + fixtures

DONE

~0.5 day

Run KONZ to completion; document the live output path, filename convention, variables, and units; stage the five Drive reference copies into the repo as the validation oracle.

Done when: a reviewer can locate a live history file from the documented path without guessing.

Outcome: `docs/data-contract.md` written against a real KONZ run. Surfaced the stream-naming mismatch that reshaped Phase 1. Fixture population from Drive is still outstanding.

PHASE 1 Rebind the data-access layer

NEXT

~1 day

The one real refactor. First fix the history-file pattern in both `data_access.py` and `run_neon_v2.py` so it matches live output at all (stream suffix, date form, path shape) while still reading the legacy-named reference copies. Then add a local reader, route it through a single source-agnostic entry point (env-var selected, local default, S3 opt-in), and remove the hardcoded S3 path. Confirm the xarray engine: `scipy` suited the NetCDF-3 fixtures but live files may be NetCDF-4. S3 functions stay available.

Done when: data loads from a live run with no S3 credentials present, asserted non-empty rather than merely not raising; legacy-named reference copies also read; the S3 path still works when explicitly selected.

PHASE 2 Hub 1 — Data Analysis

~0.5 day

Simplest Hub first. Swap the notebook's S3 calls for the new entry point, drop the mandatory credential preflight, re-run against live output.

Done when: plots and histograms render from live output and match the KONZ reference within tolerance.

PHASE 3 Hub 2 — Modeling / Kalman filter

~1 day

Point the misfit and calibration workflow at native output, confirm the filter targets GPP, validate bias / residual / R^2 and the calibrated result against the reference notebook.

Done when: misfit and calibration numbers reproduce the reference results within tolerance on at least two sites.

PHASE 4 Hub 3 — Experimentation

~1.5 days

Wire the confirmed 4-step loop: one NEON input → one perturbed copy (precipitation first — fully codified) → run CESM on both → t-test the outputs. Confirm the in-scope perturbation set.

Done when: one notebook run produces two CESM outputs and a t-test result for a precipitation perturbation, with no manual steps.

PHASE 5 Multi-site validation + hardening

~1 day

Run all three Hubs across the five sites against the reference copies, address the known Mac/Linux slowness, update the getting-started guide for the native data flow.

Done when: all three Hubs run clean on all five sites from a fresh container pull.

Open decisions to confirm with Jingyi

None block Phases 0–2; each is needed before Phase 4 closes.

DECISION	STATUS	WHY IT MATTERS
Perturbation variable scope for the MVP	OPEN	The doc is inconsistent (one line literally reads "PFT, Soil type and [] "). Precipitation-only may be enough for the MVP.
Validation tolerance vs. reference copies	OPEN	A live re-run will not be bit-identical to archived output, so "validation" needs a defined threshold.
The five site codes	OPEN	KONZ is the proven baseline; the other four need confirming.

Risks

RISK	MITIGATION
Live runs won't bit-match archived reference output	Settle a numeric tolerance before signing off Phase 2.
Compute time (KONZ ~337 s per run; Hub 3 needs two runs per site across five sites)	Budget wall-clock; consider running site validations in the background.
PFT / soil perturbations are less exercised than forcing transforms	Keep them off the MVP critical path unless Jingyi requires them.

Current status at a glance

CAPABILITY	STATUS
CESM/CTSM simulation runs to completion (KONZ)	PROVEN
Native NEON forcing pull (inherited from CTSM)	WORKING
Hub notebooks + analytics backend + perturbation tooling	EXISTS
Data contract for live output (Phase 0)	DONE
Reference copies populated into fixtures	TO DO
Hubs read from live/native output	TO DO
Multi-site validation against reference copies	TO DO
Perturbation scope, tolerance, site list	AWAITING JINGYI

Full implementation detail, file-level change list, and per-phase acceptance criteria are in the companion document `hub-integration-plan.md`.